Problem 4.25

(a)

```
rm(list=ls())
library(POE5Rdata)
#problem4.25
mod1 = lm(log(price) ~ sqft, data = collegetown)
mod2 = lm(log(price) ~ log(sqft), data = collegetown)
mod3 = lm(price ~ sqft, data = collegetown)

#(a)
summary(mod1)
```

```
Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.32528 -0.19199  0.00122  0.21678  0.86609 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,	Adjusted R-squared:  0.5408 
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

Interpret the estimated parameters:

Beta1 的 estimate 為 b1 = 4.393866，意思是當解釋變數 sqft = 0，price 的估計值為 exp^(4.393866)(千元)；Beta2 的 estimate 為 b2 = 0.036044，意思是當解釋變數 sqft 上升一單位，price 平均而言會上升 3.6044%。

```
# slope and elasticity at sample mean
slope_a = coef(summary(mod1))[[2]] * mean(collegetown$price)
elas_a = coef(summary(mod1))[[2]] * mean(collegetown$sqft)
cat(
  "(a)" ,"\n",
  "slope at the sample mean is : ",slope_a, "\n",
  "elasticity at the sample mean is : ",elas_a,"\n",
  sep=""
)
```

```
slope at the sample mean is : 9.019661
elasticity at the sample mean is : 0.9833701
```

(b)

```
#(b)
summary(mod2)
```

```
Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36864 -0.20849  0.00487  0.23204  1.21099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.04965    0.15797   12.97   <2e-16 ***
log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,	Adjusted R-squared:  0.4728 
F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

Interpret the estimated parameters:

Beta1 的 estimate 為 b1 = 2.04965，意思是當解釋變數 log(sqft) = 0，price 的期望值為 exp^(2.04965)(千元)；Beta2 的 estimate 為 b2 = 1.02483，意思是當解釋變數 sqft 上升 1%，price 平均而言會上升 1.02483%。

```
# slope and elasticity at sample mean
slope_a = coef(summary(mod2))[[2]] * (mean(collegetown$price) / mean(collegetown$sqft))
elas_a = coef(summary(mod2))[[2]]
cat(
  "(b)" ,"\n",
  "slope at the sample mean is : ",slope_a, "\n",
  "elasticity at the sample mean is : ",elas_a,"\n",
  sep=""
)
```

```
(b)
slope at the sample mean is : 9.399923
elasticity at the sample mean is : 1.024828
```

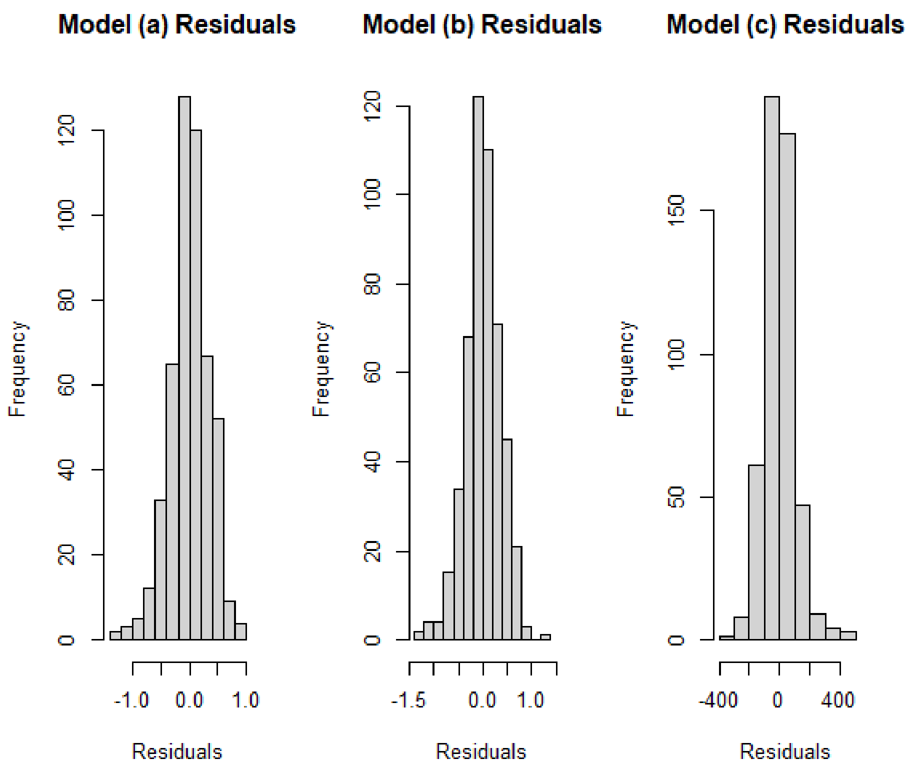
(c)

```r
#(c)
cat(
  "(c)" ,"\n",
  "R-squared value from the linear model is : ",summary(mod3)$r.squared, "\n",
  "Generalized R-squared from (b) is : ",cor(collegetown$price, exp(fitted(mod2)))^2,"\n",
  "Generalized R-squared from (c)(the linear model) is : ",cor(collegetown$price, fitted(mod3))^2,"\n",
  sep=""
)
```

```
R-squared value from the linear model is : 0.6413167
Generalized R-squared from (b) is : 0.6445084
Generalized R-squared from (c)(the linear model) is : 0.6413167
```

(d)

```r
#(d)
par(mfrow = c(1, 3))
hist(resid(mod1), main = "Model (a) Residuals", xlab = "Residuals")
hist(resid(mod2), main = "Model (b) Residuals", xlab = "Residuals")
hist(resid(mod3), main = "Model (c) Residuals", xlab = "Residuals")
cat(
  "(d)" ,"\n",
  "the Jarque-Bera statistics for (a) is : ",jarque.bera.test(resid(mod1))[[1]], "\n",
  "the Jarque-Bera statistics for (b) is : ",jarque.bera.test(resid(mod2))[[1]],"\n",
  "the Jarque-Bera statistics for (c) is : ",jarque.bera.test(resid(mod3))[[1]],"\n",
  sep=""
)
```



```
(d)
the Jarque-Bera statistics for (a) is : 26.67899
the Jarque-Bera statistics for (b) is : 14.27871
the Jarque-Bera statistics for (c) is : 221.1115
```

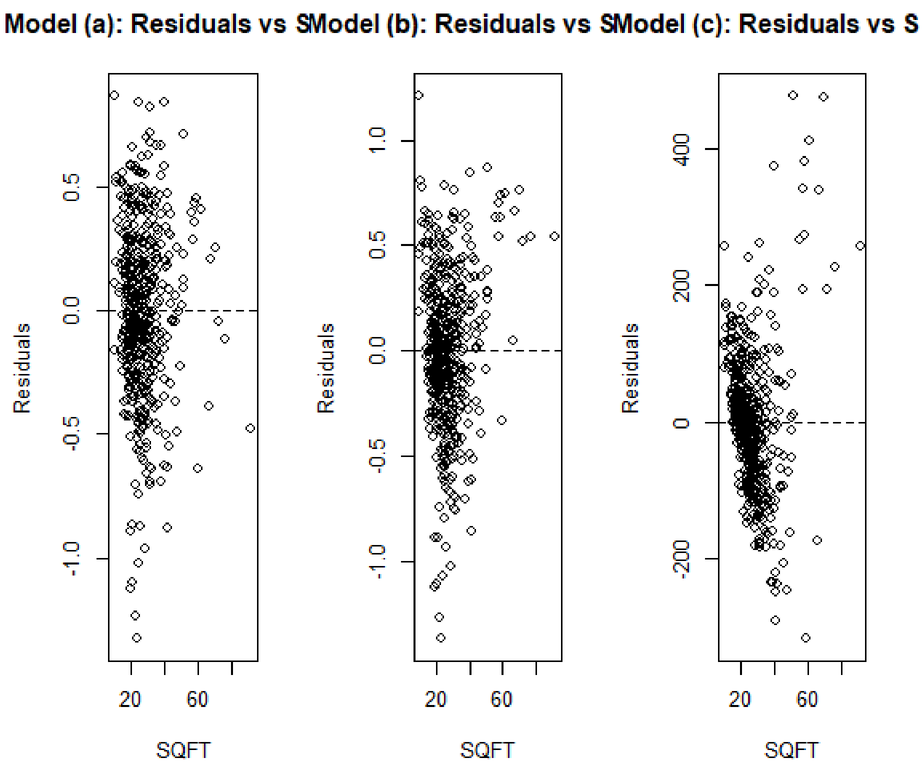
```
cat(
  "(d)" ,"\n",
  "p-value for (a) is : ",jarque.bera.test(resid(mod1))[[3]], "\n",
  "p-value for (b) is : ",jarque.bera.test(resid(mod2))[[3]],"\n",
  "p-value for (c) is : ",jarque.bera.test(resid(mod3))[[3]],"\n",
  sep=""
)
```

```
(d)
p-value for (a) is : 1.609651e-06
p-value for (b) is : 0.0007932632
p-value for (c) is : 0
```

從上圖最後 p-value 的圖可以發現，a,b,c 三個模型的 residuals 之 JB test 之 p-value 都小於 alpha = 0.05，因此可以拒絕虛無假設，也就是 residual 不完全符合常態分配(但 model2 的 JB 值最小，為三者之中更偏向常態分者)。

(e)

```
par(mfrow = c(1, 3))
plot(collegetown$sqft, resid(mod1),
     main = "Model (a): Residuals vs SQFT",
     xlab = "SQFT", ylab = "Residuals")
abline(h = 0, lty = 2)
plot(collegetown$sqft, resid(mod2),
     main = "Model (b): Residuals vs SQFT",
     xlab = "SQFT", ylab = "Residuals")
abline(h = 0, lty = 2)
plot(collegetown$sqft, resid(mod3),
     main = "Model (c): Residuals vs SQFT",
     xlab = "SQFT", ylab = "Residuals")
abline(h = 0, lty = 2)
```



Residuals 的分布基本上符合模型假設，也就是在 y = 0 上下擺動(期望值為 0)

(f)

```
#(f)
newhouse <- data.frame(sqft = 27) #單位是千美元
cat(
  "(f)" ,"\n",
  "predict value for model (a) is : ",exp(predict(mod1, newhouse))*1000, "\n",
  "predict value for model (b) is : ",exp(predict(mod2, newhouse))*1000,"\n",
  "predict value for model (c) is : ",predict(mod3, newhouse)*1000,"\n",
  sep=""
)
```

```
(f)
predict value for model (a) is : 214233.6
predict value for model (b) is : 227538.6
predict value for model (c) is : 246455.7
```

(g)

```r
#(g)
pi_a = exp(predict(mod1, newdata = newhouse, interval = "prediction", level = 0.95))
pi_b = exp(predict(mod2, newdata = newhouse, interval = "prediction", level = 0.95))
pi_c = predict(mod3, newdata = newhouse, interval = "prediction", level = 0.95)
cat(
  "(g): 95%PI" ,"\n",
  "model (a) log-linear: ",pi_a[[2]],",",pi_a[[3]], "\n",
  "model (b) log-log: ",pi_b[[2]],",",pi_b[[3]],"\n",
  "model (c) linear : ",pi_c[[2]],",",pi_c[[3]],"\n",
  sep=""
)
```

```
(g): 95%PI
model (a) log-linear: 109.7685,418.1167
model (b) log-log: 111.1406,465.8406
model (c) linear : 44.27727,448.6341
```

(h)

Linear model 表現較不好，因為 JB's statistics 太高，且判定係數也略小於 log-log model。

另外，log-linear 以及 log-log 兩者都可以當作選擇的模型來使用，前者有較高的判定係數，後者的 JB's statistics 比較小。